

State of California
CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
SANTA ANA REGION

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ORDER NO. R8-2015-0023

**WASTE DISCHARGE REQUIREMENTS
AND
MASTER RECLAMATION PERMIT
FOR
EL TORO WATER DISTRICT
WATER RECYCLING PLANT
ORANGE COUNTY**

The following Discharger is subject to waste discharge and producer/user reclamation requirements as set forth in this Order:

Table 1. Discharger Information

Discharger/Operator	El Toro Water District
Name of Facility	Water Recycling Plant
Facility Address	23542 Moulton Parkway, Laguna Woods, CA 92654
	Orange County

The discharge by the El Toro Water District from the discharge point identified below is subject to waste discharge requirements as set forth in this Order:

Table 2. Discharge Locations and Recycled Water Use Areas

Discharge Point	Effluent Description	Discharge Point (Latitude)	Discharge Point (Longitude)	Receiving Water
DP-001 Recycled water	Up to 4 mgd of tertiary treated and disinfected	33°37'08"N	117°43'37"W	Irvine Groundwater Management Zone

Table 3. Administrative Information

This Order was adopted by the Regional Water Board on:	June 19, 2015
This Order shall become effective on:	June 19, 2015

IT IS HEREBY ORDERED that this Order supersedes and rescinds Order No. 94-03, except for enforcement purposes, and, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, El Toro Water District shall comply with the requirements set forth in this Order.

I, Kurt V. Berchtold, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Santa Ana Region, on June 19, 2015.


for Kurt V. Berchtold, Executive Officer

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I. DISCHARGER/FACILITY INFORMATION

The following Discharger is subject to waste discharge and producer/user reclamation requirements as set forth in this Order:

Table 4. Discharger/Facility Information

Discharger	El Toro Water District
Address	24251 Los Alisos Boulevard, Lake Forest, CA 92630
Facility	Water Recycling Plant
Facility Address	23542 Moulton Parkway, Laguna Woods, CA 92654
Facility Contact	Mark Pade, (949) 837-7050, extension 103, mpade@etwd.com
Type of Facility	POTW
Facility Design Flow - secondary treatment	6 mgd secondary treated
Facility Design Flow -tertiary treatment	4 mgd tertiary treated and disinfected

II. FINDINGS

The California Regional Water Quality Control Board, Santa Ana Region (hereinafter Regional Water Board), finds:

- A. Background:** El Toro Water District (hereinafter Discharger or ETWD) owns and operates a sanitary sewer collection system, a Water Recycling Plant (hereinafter, the Facility), and a recycled water distribution system. ETWD's service area includes portions the Cities of Lake Forest, Laguna Hills, Aliso Viejo, Mission Viejo, and all of the City of Laguna Woods. The Discharger owns the Facility and is currently discharging from the Facility pursuant to waste discharge and producer/user water reclamation requirements, Order No. 94-3. Order 94-3 was issued to the South Orange County Reclamation Authority (SOCRA, predecessor to South Orange County Wastewater Authority, SOCWA). ETWD is a member of SOCWA. ETWD's service area extends into the San Diego and the Santa Ana Regional Water Board jurisdictions. The Santa Ana Water Board issued Order No. 94-3 to regulate tertiary treated recycled water discharges within the Santa Ana Region. ETWD is proposing to reclaim, and distribute up to 4 million gallon per day (mgd) of tertiary treated disinfected wastewater within its service area. When reclaimed water demand is low, ETWD proposes to discharge up to 6 mgd of secondary treated un-disinfected wastewater through SOCWA's Aliso Creek ocean outfall under the San Diego Regional Water Board's Order No. R9-2012-0013, NPDES No. CA0107611, issued to SOCWA.

The Discharger recently completed improvements to the treatment capabilities of the Facility by constructing a new tertiary treatment and recycled water distribution system

that consists of two rotating-disk cloth-media filters, tertiary effluent transfer pumps, a recycled water storage tank with integrated chlorine contact basin (dual purpose), a recycled water pump station, an alum/poly blend storage and feed station, and approximately 19 miles of pipelines for distribution of recycled water, among other improvements. Although the Discharger has designed the tertiary treatment system to produce up to 4.0 mgd of tertiary-2.2 recycled water, initially, the installed tertiary treatment capacity is 3.7 mgd and, in the future, the tertiary treatment capacity can be expanded to 4 mgd by adding two disks to each filtration unit. Each filtration unit can be equipped with a maximum of 12 disks and currently each unit only has 8 disks. As of February 2015, the Discharger started the distribution of tertiary treated and disinfected recycled water to new use areas connected to its distribution pipeline and expects to distribute up to 4.0 mgd of recycled water in the future as new use areas are connected, tested and approved. The recycled water use areas overlie the Irvine Groundwater Management Zone.

- B. Facility Description:** The Discharger owns and operates the Water Recycling Plant and its wastewater treatment processes include: preliminary treatment that consist of influent flow metering, mechanical bar-screening, aerated grit chamber, and fine screening (roto-strainers); influent flow equalization; odor control with inorganic media filter beds; secondary treatment that consists of an activated sludge process with limited nutrient removal (nitrification/denitrification) and one secondary effluent holding pond; tertiary treatment system that uses rotating-disk cloth filters and disinfection with chlorine (chlorine contact basin). The secondary effluent holding pond covers approximately 2.5 acres and is used to store, temporarily, secondary effluent under emergency conditions when the Facility is not able to dispose of the secondary effluent through the ocean outfall. Normally, the secondary effluent stored in the holding pond is gravity fed to the wet-well of the ocean outfall pump station. The Facility is designed to treat up to 6.0 mgd of wastewater to secondary treatment standards and up to 4.0 mgd of tertiary-2.2 recycled water. Also, the Discharger has constructed a recycled water distribution system composed of a storage tank, pump station, and approximately 19 miles of pipeline. Attachment B is a location and vicinity map for this facility. Attachment C is a flow schematic of the treatment systems.
- C. Legal Authorities:** This Order serves as Waste Discharge Requirements (WDRs) pursuant to Article 4, Chapter 4, Division 7 of the California Water Code (commencing with section 13260). Also, this Order serves as a master reclamation permit pursuant to Section 13523.1 of Article 4, Chapter 7, Division 7 of the California Water Code.
- D. Background and Rationale for Requirements:** The requirements in this Order are based on information submitted as part of the application and other available information. Attachment F, which contains background information and rationale for requirements in this Order, is hereby incorporated into this Order, and thus constitutes a part of the Findings for this Order. Attachments A, B, C, E, G and H are also incorporated into this Order.

E. Statewide General Waste Discharge Requirements for Sanitary Sewer Systems:

The State Water Resources Control Board issued the Statewide General Waste Discharge Requirements for Sanitary Sewer Systems, Order No. 2006-0003-DWQ (General Order) on May 2, 2006, as amended by Order No. WQ 2008-0002-EXEC, requiring public agencies that own sanitary sewer systems comprised of more than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order requires agencies to develop sanitary sewer management plans and report all sanitary sewer overflows. The Discharger is currently enrolled under the General Order.

F. Monitoring and Reporting: Water Code section 13267 authorizes the Regional Board to require technical and monitoring reports. The Monitoring and Reporting Program establishes monitoring and reporting requirements to implement federal and State requirements. This Monitoring and Reporting Program is provided in Attachment E of this Order.

G. CEQA Compliance: This Order updates requirements for an existing facility. Furthermore, the Discharger prepared appropriate environmental documents and filed Notices of Determination related to the treatment plant upgrades and the recycled water distribution system. Regional Board staff reviewed the environmental documents prepared for the Facility and recycled water distribution systems and determined that if the requirements specified herein are complied with, there should not be any significant water quality impacts from the permitted discharges. See the Fact Sheet, Attachment F, for details.

H. Notification of Interested Parties: The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe Waste Discharge Requirements and a Master Reclamation Permit for this project and has provided them with an opportunity to submit their written comments and recommendations. Details of the notification are provided in the Fact Sheet (Attachment F) of this Order.

I. Consideration of Public Comment: The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the public meeting are provided in the Fact Sheet (Attachment F) of this Order.

III. PROHIBITIONS

- A.** The use of recycled water shall be limited to treated effluent that meets the conditions and requirements specified in Section IV.A.
- B.** The unauthorized discharge of wastewater and/or of recycled water at a location or in a manner different from those described in this Order is prohibited.
- C.** The bypass or overflow of untreated wastewater or wastes to surface waters or surface water drainage courses is prohibited.

- D. The discharge of any substances in concentrations toxic to animal or plant life in the affected receiving water is prohibited.
- E. The discharge of any radiological, chemical, or biological warfare agent or high level radiological waste is prohibited.
- F. The distribution and use of recycled water prior to authorization by the California State Water Resources Control Board, Division of Drinking Water, is prohibited.

IV. RECLAMATION SPECIFICATIONS

A. Reclamation Specifications: Discharge Point 001 (DP-001)

The Discharger shall maintain compliance with the following limitations for recycled water produced and distributed for landscape irrigation or other similar uses. Compliance is to be measured at monitoring location REC-001 or other approved monitoring locations where representative samples of recycled water can be obtained for laboratory testing and analysis, as described in the attached Monitoring and Reporting Program (Attachment E). The Discharger may submit for approval by the Executive Officer other monitoring location(s) not specified herein where representative samples of recycled water could be obtained for laboratory testing and analysis.

1. **Physical/Biological Limitations:** The Discharger shall maintain compliance with the following recycled water limitations at REC-001:

Table 5. Recycled Water Limitations at REC-001

Parameter	Units	Limitations	
		Average Monthly	Average Weekly
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	20	30
Total Suspended Solids	mg/L	20	30

2. **TDS Limitations:** The 12-month flow weighted running average total dissolved solids concentration of the discharge at Discharge Point 001 shall not exceed the 910 mg/l unless:
 - (1) The Discharger demonstrates to the satisfaction of the Regional Water Board's Executive Officer that:
 - (a) Discharges in excess of the TDS limits are due to the quality of water supply sources utilized in the Discharger's service area, and that all reasonable steps, as

agreed upon by the Executive Officer, have been taken to ensure that the best quality supplies are obtained and utilized in the Discharger's service area; and/or

(b) Discharges in excess of the TDS limits are due to chemical additions in the treatment process needed to meet waste discharge requirements, and the Discharger has taken all steps to optimize chemical additions so as to minimize the increases; and

(2) The Discharger implements an acceptable plan to offset discharges in excess of the TDS limits.

3. **Recycled Water Specifications:** The recycled water shall all times be adequately oxidized disinfected tertiary treated recycled water, which is a filtered and subsequently disinfected wastewater that meets the following limitations:

(1) Turbidity of the filtered effluent

When filtration¹ is through natural undisturbed soils or a bed of filter media, the turbidity of the filtered wastewater shall not exceed any of the following:

- (a) Average of 2 Nephelometric Turbidity Unit (NTU) within any 24-hour period;
- (b) 5 NTU more than 5 percent of the time in any 24-hour period; and
- (c) 10 NTU at any time.

(2) Disinfected wastewater shall meet the following:

- (a) The median concentration of total coliform bacteria in the disinfected effluent shall not exceed a Most Probable Number (MPN) of 2.2 per 100 milliliters (ml) utilizing the bacteriological results of the last seven days for which analysis has been completed.
- (b) The number of total coliform organism shall not exceed an MPN of 23 total coliform bacteria per 100 ml in more than one sample in any calendar month.
- (c) No total coliform sample shall exceed an MPN of 240 total coliform bacteria per 100 ml.
- (d) When chlorine disinfection process is utilized following filtration, a CT (the product of total chlorine residual and modal contact time² measured at the same point) value of not less than 450 milligram-minutes per liter at all times

¹ For recycled water use, other acceptable filtration technology that complies with Title 22 of the California Code of Regulations and approved by the California SWRCB's DDW may be used. Compliance determination will be based on California SWRCB's DDW guidance.

² Modal contact time shall be calculated daily based on the minimum one-hour average value in a 24-hour period.

with a modal contact time of at least 90 minutes, based on peak dry weather design flow shall be provided.

- (e) When a disinfection process combined with the filtration process is utilized, the combined process shall demonstrate inactivation and/or removal of 99.999 percent of the plaque-forming units of F-specific bacteriophage MS2, or polio virus in the wastewater. A virus that is at least as resistant to disinfection as polio virus may be used for purposes of the demonstration. The facility must be operated and maintained in accordance with a California State Water Resources Control Board's Division of Drinking Water³ (SWRCB's DDW), approved operations plan, which is part of the Title 22 Engineering Report. The operations plan included in the Title 22 Engineering Report shall become an enforceable part of this Order.
- 4. The pH of the recycled water shall at all times be within the range of 6.0 to 9.0 pH units.
 - 5. The Discharger shall prepare an Engineering Report, conforming to Section 60323, Article 7, Chapter 3, Division 4, Title 22 of the California Code of Regulations, and at a programmatic level consistent with the planned distribution and uses of recycled water represented in this Order. The Programmatic Engineering Report shall be submitted to the California SWRCB's DDW and the Regional Water Board and shall be approved by the California SWRCB's DDW prior to delivery of recycled water to any user.
 - 6. Prior to the delivery of recycled water to any schools, hospitals, or other unique use sites, the Discharger shall submit to the California SWRCB's DDW and the County Environmental Health Department for review and approval a report containing the information listed in paragraph IV.A.11., below.
 - 7. The Discharger shall be responsible for assuring that recycled water is delivered and utilized in conformance with this Order and the recycling criteria contained in Title 22, Division 4, Chapter 3, Sections 60301 through 60355, California Code of Regulations. The Discharger shall conduct periodic inspections of the facilities of the recycled water users to monitor compliance by the users with this Order.
 - 8. The Discharger shall establish and enforce Rules and Regulations for Recycled Water users, governing the design and construction of recycled water use facilities and the use of recycled water in accordance with the uniform statewide recycling criteria established pursuant to California Water Code Section 13521.
 - a. Use of recycled water by the Discharger shall be consistent with its Rules and Regulations for Recycled Water Use.

³

Before July 1, 2014 the California Department of Public Health.

- b. Any revisions made to the Rules and Regulations shall be subject to the review of the Regional Water Board, the California SWRCB's DDW, and the County Environmental Health Department. The revised Rules and Regulations or a letter certifying that the Discharger's Rules and Regulations contain the updated provisions in this Order, shall be submitted to the Regional Water Board within 60 days of adoption of this Order by the Regional Water Board.
9. The Discharger shall conduct periodic inspections of recycled water reuse sites to monitor compliance with the Discharger's Rules and Regulations for Recycled Water Use and the uniform statewide reclamation criteria established pursuant to California Water Code Section 13521.
10. The storage, delivery, or use of recycled water shall not individually or collectively, directly or indirectly, result in a pollution or nuisance, or adversely affect water quality, as defined in the California Water Code.
11. The Discharger shall maintain and make available upon request by the Regional Water Board the following information for any recycled water users:
 - a. The average number of persons estimated to be served at each use area on a daily basis.
 - b. The specific boundaries of the proposed use area including a map showing the location of each facility, drinking water fountain, and impoundment to be used.
 - c. The person or persons responsible for operation of the recycled water system at each use area.
 - d. The specific use to be made of the recycled water at each use area.
 - e. The methods to be used to assure that the installation and operation of the recycled water system will not result in cross connections between the recycled water and potable water piping systems. This shall include a description of the pressure, dye or other test methods to be used to test the system.
 - f. Plans and specifications which include following:
 - (1) Proposed piping system to be used.
 - (2) Pipe locations of both the recycled and potable water systems.
 - (3) Type and location of the outlets and plumbing fixtures that will be accessible to the public.
 - (4) The methods and devices to be used to prevent backflow of recycled water into the potable water system.
 - (5) Plan notes relating to specific installation and use requirements.
12. The Discharger shall require each user to designate an on-site supervisor responsible for the operation of the recycled water distribution system within the use area. The supervisor shall be responsible for complying with this Order, prevention of potential hazards, the installation, operation and maintenance of the distribution system as

approved by California SWRCB's DDW (previously the California Department of Public Health).

V. RECEIVING WATER LIMITATIONS AND SPECIFICATIONS

A. Surface Water Limitations

The discharge of waste or use of recycled water shall not cause any surface waters to be degraded, cause water quality objectives to be exceeded, unreasonably affect beneficial uses, or cause a condition of pollution or nuisance.

B. Groundwater Limitations

The discharge of waste or use of recycled water shall not cause the underlying groundwater to be degraded, cause water quality objectives to be exceeded, unreasonably affect beneficial uses, or cause a condition of pollution or nuisance.

VI. PROVISIONS

A. General Provisions

1. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this facility, may subject the Discharger to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Discharger to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.
2. In the event the Discharger does not comply or will be unable to comply for any reason with any prohibition, discharge limitation, or receiving water limitation of this Order, which pose an immediate threat to human health or the environment, the Discharger shall notify the Regional Water Board by telephone at (951) 782-4130 or by email to info8@waterboards.ca.gov within 24 hours of having knowledge of such noncompliance, and shall confirm this notification in the monthly Self-Monitoring Report, unless the Regional Water Board waives confirmation or requires, orally or in writing, a written notification within five business days. The written notification shall state the nature, time, duration, and cause of noncompliance, and shall describe the measures being taken to remedy the current noncompliance and, prevent recurrence including, where applicable, a schedule of implementation. Other noncompliance requires written notification with the details discussed above with the next monitoring report.
3. Neither the treatment nor the discharge of pollutants shall create a pollution, contamination, or nuisance as defined by Section 13050 of the CWC.

4. The Discharger shall take all reasonable steps to minimize or correct any adverse impact on the environment resulting from noncompliance with this Order, including such accelerated or additional monitoring as may be necessary to determine the nature and impact of the noncomplying discharge.
5. This Order may be modified, revoked and reissued, or terminated for cause including, but not limited to, the following.
 - a. Violation of any terms or conditions of this Order;
 - b. Obtaining this Order by misrepresentation or failure to disclose fully all relevant facts.
6. The Discharger shall file with the Regional Water Board a Report of Waste Discharge at least 140 days before making any material change in the character, location, or volume of the discharge. A material change includes, but is not limited to, the following:
 - a. Adding a major industrial waste discharge to a discharge of essentially domestic sewage, or adding a new process or product by an industrial facility resulting in a change in the character of the waste.
 - b. Significantly changing the disposal method or location, such as changing the disposal to another drainage area or water body.
 - c. Significantly changing the method of treatment.
 - d. Increasing the treatment plant design capacity beyond that specified in this Order.
7. The provisions of this Order are severable, and if any provision of this Order, or the application of any provision of this Order to any circumstance, is held invalid, the application of such provision to other circumstances, and the remainder of this Order, shall not be affected thereby.
8. The Discharger shall maintain a copy of this Order at the site so that it is available to site operating personnel at all times. Key operating personnel shall be familiar with its content.
9. The Discharger shall optimize chemical additions needed in the treatment process to meet waste discharge requirements so as to minimize total dissolved solid increases in the treated wastewater.
10. Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Regional Water Board's Executive Officer.
11. In the event of any change in control or ownership of land or waste discharge facility presently owned or controlled by the Discharger, the Discharger shall notify the

succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to the Regional Water Board.

12. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

13. The Regional Water Board and other authorized representatives shall be allowed:

- a. Entry upon premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order;
- b. Access to copy any records that are kept under the conditions of the Order;
- c. To inspect any facility, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order; and
- d. To photograph, sample and monitor for the purpose of assuring compliance with this Order, or as otherwise authorized by the Water Code.

B. Monitoring and Reporting Program Requirements

The Discharger shall comply with the Monitoring and Reporting Program, and future revisions thereto, in Attachment E of this Order. This monitoring and reporting program may be modified by the Executive Officer at any time during the term of this Order, and may include a reduction or an increase in the number of parameters to be monitored, the frequency of the monitoring or the number and size of samples to be collected.

C. Special Provisions

1. Special Studies, Technical Reports and Additional Monitoring Requirements (not applicable)

2. Best Management Practices and Pollution Prevention –

- a. Pollutant Minimization Program (not applicable)

3. Construction, Operation and Maintenance Specifications

- a. The Discharger's wastewater treatment plant shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to Title 23, Division 3, Chapter 26, California Code of Regulations.
- b. The Discharger shall provide safeguards to assure that should there be reduction, loss, or failure of electric power, the Discharger will comply with the requirements of this Order.
- c. The Discharger shall update as necessary, the "Operation and Maintenance Manual (O&M Manual)," which it has developed for the treatment facility to

conform to latest plant changes and requirements. The O&M Manual shall be readily available to operating personnel onsite. The O&M Manual shall include the following:

- (1) Description of the treatment plant's organizational chart showing the number of employees, duties and qualifications and plant attendance schedules (daily, weekends and holidays, part-time, etc). The description should include documentation that the personnel are knowledgeable and qualified to operate the treatment facility so as to achieve the required level of treatment at all times.
- (2) Detailed description of safe and effective operation and maintenance of treatment processes, process control instrumentation and equipment.
- (3) Description of laboratory and quality assurance procedures.
- (4) Process and equipment inspection and maintenance schedules.
- (5) Description of safeguards to assure that, should there be reduction, loss, or failure of electric power, the Discharger will be able to comply with requirements of this Order.
- (6) Description of preventive (fail-safe) and contingency (response and cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. These plans shall identify the possible sources of accidental discharges, untreated or partially treated waste bypass, and polluted drainage (such as: loading and storage areas, power outage, waste treatment unit failure, process equipment failure, tank and piping failure, etc.).

4. Special Provisions for Municipal Facilities (POTWs Only)

- a. Sewer Collection System Requirements: The Discharger's collection system is part of the system that is subject to this Order. As such, the Discharger must properly operate and maintain its collection system (40 C.F.R. § 122.41(e)). The Discharger must report any non-compliance (40 C.F.R. § 122.41(l)(6) and (7)) and mitigate any discharge from the Discharger's collection system in violation of this Order (40 C.F.R. § 122.41(d)).

Furthermore, the Statewide General Waste Discharge Requirements for Sanitary Sewer Systems, Order No. 2006-0003-DWQ (General Order), contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. While the Discharger must comply with both the General Order and this Order, the General Order more clearly and specifically stipulates requirements for operation and maintenance and for reporting and mitigating sanitary sewer overflows. The Discharger and other governmental agencies that are discharging wastewater into the Facility are required to obtain coverage under the General Order.

- b. Sludge Disposal Requirements

- (1) Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with State Water Board and CalRecycle's joint regulations (Title 27) of the California Code of Regulations and approved by the Regional Water Board's Executive Officer.
- (2) The use and disposal of biosolids shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.
- (3) Any proposed change in biosolids use or disposal practice from a previously approved practice should be reported to the Executive Officer and EPA Regional Administrator at least 90 days in advance of the change.
- (4) The Discharger shall take all reasonable steps to minimize or prevent any discharge or biosolids use or disposal that has the potential of adversely affecting human health or the environment.

c. Pretreatment Program Requirements (Not applicable)

VII. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in Section IV of this Order will be determined as specified below:

A. General

Compliance with effluent limitations for pollutants shall be determined using sample reporting protocols defined in the MRP and Attachment A of this Order. For purposes of reporting and administrative enforcement by the Regional and State Water Boards, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the minimum level (ML).

B. Average Monthly Effluent Limitation (AMEL)

If the average (or when applicable, the median determined by subsection F below for multiple sample data) of daily discharges over a calendar month exceeds the AMEL for a given parameter, this will represent a single violation, though the Discharger will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month). If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the AMEL, the Discharger will be considered out of compliance for that calendar month. The Discharger will only be considered out of compliance for days when the discharge occurs. For any one calendar month during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar month.

C. Average Weekly Effluent Limitation (AWEL)

If the average (or when applicable, the median for multiple sample data, see subsection F) of daily discharges over a calendar week exceeds the AWEL for a given parameter, this will represent a single violation, though the Discharger may be

considered out of compliance for each day of that week for that parameter, resulting in 7 days of non-compliance. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the AWEL, the Discharger may be considered out of compliance for that calendar week. The Discharger will only be considered out of compliance for days when the discharge occurs. For any one calendar week during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar week.

D. Maximum Daily Effluent Limitation (MDEL)

If a daily discharge or when applicable, the median determined by subsection F above for multiple sample data of a daily discharge exceeds the MDEL for a given parameter, the Discharger will be considered out of compliance for that parameter for that 1 day only within the reporting period. For any 1 day during which no sample is taken, no compliance determination can be made for that day.

E. 12-Month Running Average

Compliance with the 12-month flow weighted running average limits under Reclamation Specification IV.A. 2. shall be determined by dividing the sum of the products of the average concentration of the recycled water for each of the previous 12 months multiplied by the volume of recycled water discharged/used over, or tributary to, each groundwater management zone during each month for the previous 12 months, by the total volume discharged/used over the groundwater management zone during the previous 12 months. (Sum of monthly concentrations multiplied by monthly flows for previous 12 months divided by sum of monthly flows for previous 12 months.)

F. Multiple Sample Data

When determining compliance with an AMEL for toxic pollutants and more than one sample result is available in a month, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:

1. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
2. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

G. Turbidity Limitations

The Discharger shall be considered in compliance with Discharge Specifications IV.A.3.(1), if the following conditions are met. If the Discharger is using a properly operating backup turbidimeter, the reading of the backup turbidimeter shall be considered in determining whether there has been an actual noncompliance:

1. There are no excursions above the limits specified in Discharge Specifications IV.A.3.(1)(a) through IV.A.3.(1)(c);
2. The apparent exceedance was caused by interference with, or malfunction of, the monitoring instrument.

H. Compliance Determination

Compliance determinations shall be based on available analyses for the time interval associated with the effluent limitation. Where only one sample analysis is available in a specified time interval (e. g., monthly or weekly average), that sample shall serve to characterize the discharge for the entire interval. If quarterly sample results show noncompliance with the average monthly limit and that sample result is used for compliance determinations for each month of the quarter, then three separate violations of the average monthly limit shall be deemed to have occurred.

ATTACHMENT A – DEFINITIONS

Arithmetic Mean (μ), also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

$$\text{Arithmetic mean} = \mu = \Sigma x / n$$

where: Σx is the sum of the measured ambient water concentrations, and
 n is the number of samples.

Average Monthly Effluent Limitation (AMEL): the highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

Average Weekly Effluent Limitation (AWEL): the highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

Best Management Practices (BMPs) are methods, measures, or practices designed and selected to reduce or eliminate the discharge of pollutants to surface waters from point and nonpoint source discharges including storm water. BMPs include structural and non-structural controls, and operation and maintenance procedures, which can be applied before, during, and/or after pollution producing activities.

Bioaccumulative pollutants are those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

Carcinogenic pollutants are substances that are known to cause cancer in living organisms.

Daily Discharge: Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

Detected, but Not Quantified (DNQ) are those sample results less than the RL, but greater than or equal to the laboratory's MDL.

Estimated Chemical Concentration is the estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

Existing Discharger means any discharger that is not a new discharger. An existing discharger includes an "increasing discharger" (i.e., an existing facility with treatment systems in place for its current discharge that is or will be expanding, upgrading, or modifying its existing permitted discharge after the effective date of this Policy).

Infeasible means not capable of being accomplished in a successful manner within a reasonable period of time, taking into account economic, environmental, legal, social, and technological factors.

Inland Surface Waters are all surface waters of the State that do not include the ocean, enclosed bays, or estuaries.

Instantaneous Maximum Effluent Limitation: the highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

Instantaneous Minimum Effluent Limitation: the lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

Maximum Daily Flow is the maximum flow sample of all samples collected in a calendar day.

Maximum Daily Effluent Limitation (MDEL) means the highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

MEC: Maximum Effluent Concentration.

Median is the middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements (n) is odd, then the median = $X_{(n+1)/2}$. If n is even, then the median = $(X_{n/2} + X_{(n/2)+1})/2$ (i.e., the midpoint between the $n/2$ and $n/2+1$).

Method Detection Limit (MDL) is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in 40 CFR 136, Appendix B, revised as of May 14, 1999.

Minimum Level (ML) is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific

analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

Not Detected (ND) are those sample results less than the laboratory's MDL.

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

Pollutant Minimization Program (PMP) means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost-effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to CWC Section 13263.3(d), shall be considered to fulfill the PMP requirements. The following reporting protocols and definitions are used in determining the need to conduct a Pollution Minimization Program (PMP). Reporting protocols in the Monitoring and Reporting Program, Attachment E, Section X.B.4 describe sample results that are to be reported as Detected but Not Quantified (DNQ) or Not Detected (ND). Definitions for a Minimum Level (ML) and Method Detection Limit (MDL) are provided in Attachment A. A Reporting Level (RL) is the ML associated with an analytical method selected by the Discharger that is authorized for monitoring effluent limitations under this Order.

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code Section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the SWRCB or RWQCB.

Process Optimization means minor changes to the existing facility and treatment plant operations that optimize the effectiveness of the existing treatment processes.

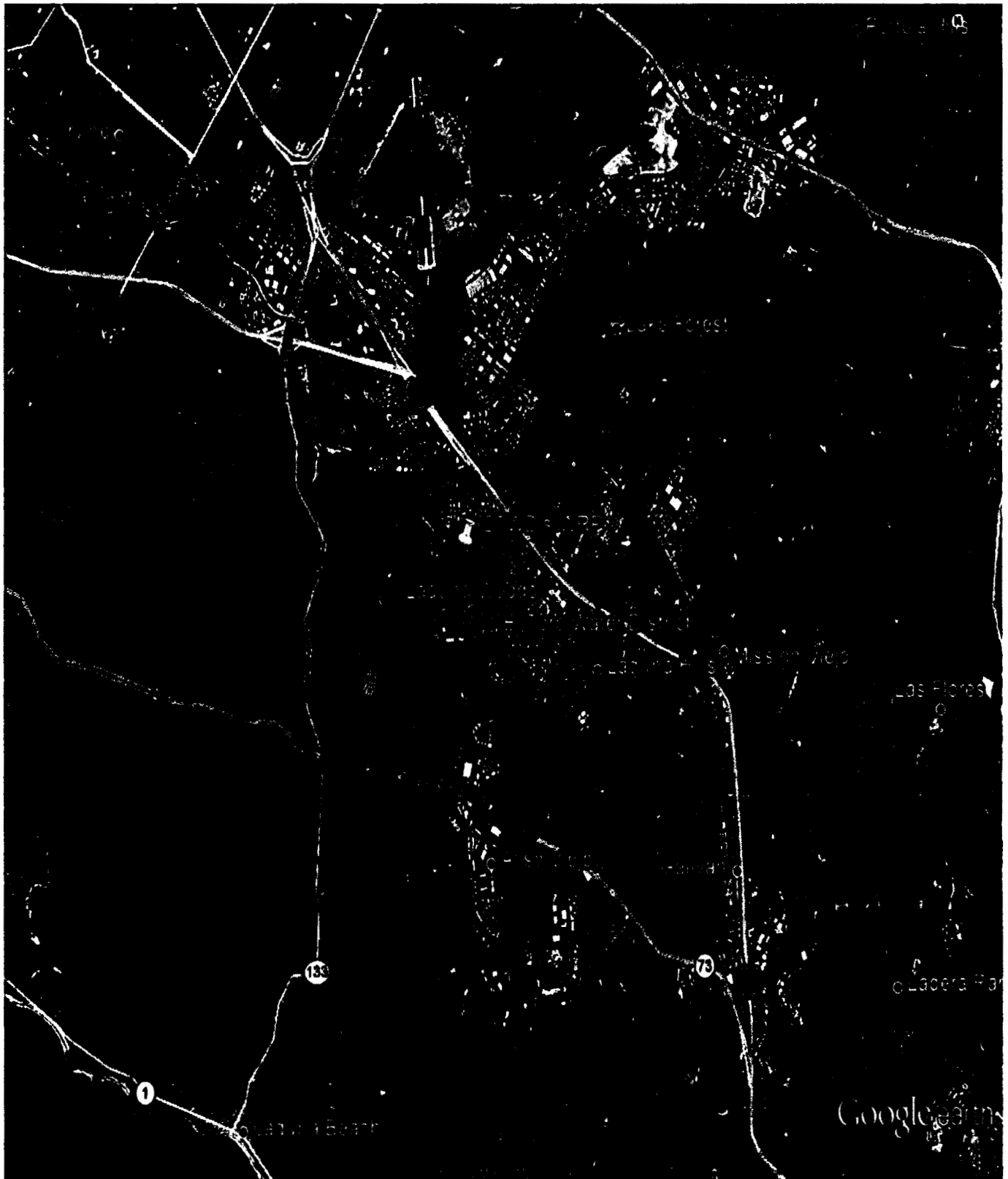
Public Entity includes the federal government or a state, county, city and county, city, district, public authority, or public agency.

Reporting Level (RL) is the ML corresponding to an approved analytical method for reporting a sample result that is selected either from Appendix 4 of the SIP by the Regional Water Board in accordance with Section 2.4.2 of the SIP or established in accordance with Section 2.4.3 of the SIP. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

Source of Drinking Water is any water designated as municipal or domestic supply (MUN) in a RWQCB basin plan.

12-Month Running Average Effluent Limitation (12-MRAEL): the highest allowable average of monthly discharges over last twelve months, calculated as the sum of all monthly discharges measured during last twelve months divided by the number of monthly discharges measured during that time period.

Attachment B1 – ETWD's WRP Location Map

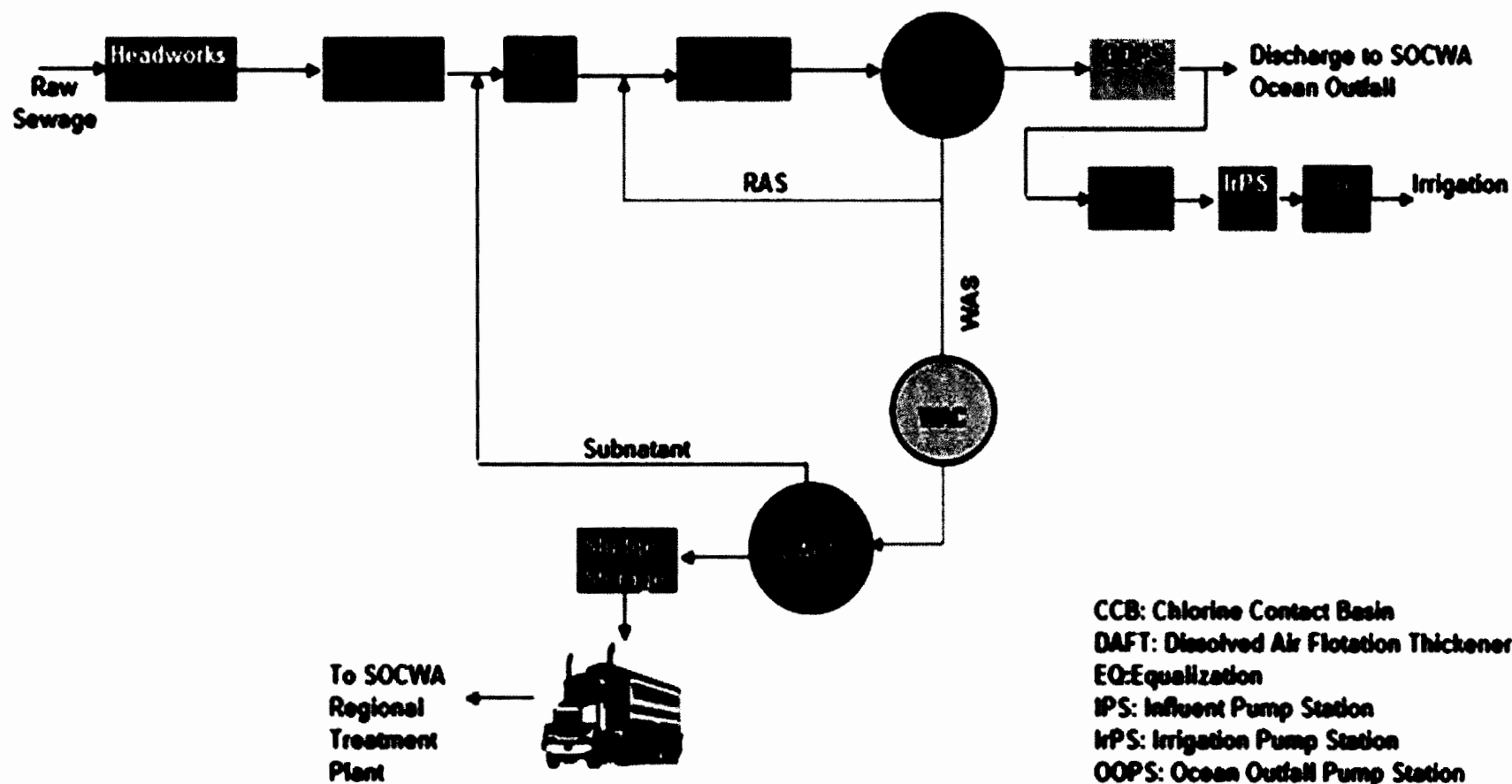


Attachment B2 - Vicinity Map of WRP



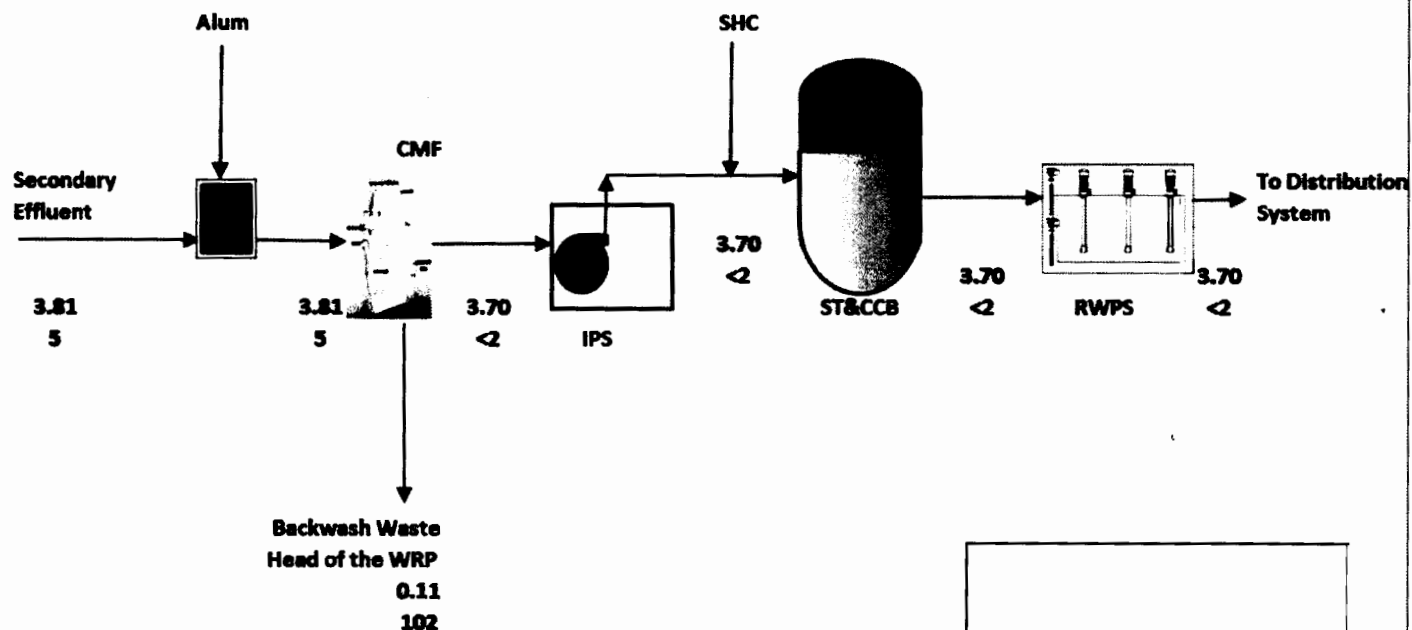
ATTACHMENT C – FLOW SCHEMATIC

Secondary Recycled Water System Schematic



CCB: Chlorine Contact Basin
 DAFT: Dissolved Air Flotation Thickener
 EQ: Equalization
 IPS: Influent Pump Station
 IrPS: Irrigation Pump Station
 OOPS: Ocean Outfall Pump Station
 RAS: Return Activated Sludge
 Scre: Screening
 SC: Secondary Clarifiers
 WAC: Waste Activated Sludge Cell
 WAS: Waste Activated Sludge

Tertiary Treated and Disinfected Recycled Water System Schematic



Legend

Flow as mgd

Turbidity as NTU

CMF: Cloth Media Filtration

IPS: Intermediate Pump Station

SHC: Sodium Hypochlorite

ST&CCB: Storage Tank with Integrated Chlorine Contact Basin

RWPS: Recycled Water Pump Station

Order No. R8-2015-0023
El Toro Water District
Water Recycling Plant

ATTACHMENT D

There is no Attachment D for this Order

Attachment E – Monitoring and Reporting Program

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ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

California Water Code Section 13267 authorizes the Regional Water Board to require technical and monitoring reports. This MRP establishes monitoring and reporting requirements that implement the federal and California regulations.

I. GENERAL MONITORING PROVISIONS

A. General Monitoring Provision

1. All sampling, sample preservation, and analytical procedures shall be in accordance with the current approved edition of "*Standard Methods for the Examination of Water and Wastewater*" (American Public Health Association) and/or 40 CFR Part 136 approved methods unless otherwise specified by the Executive Officer of the Regional Water Board.
2. Chemical, bacteriological, and bioassay analyses shall be conducted at a laboratory certified for such analyses by the California State Water Resources Control Board's Division of Drinking Water in accordance with the provision of Water Code Section 13176, and must include quality assurance/quality control data with their reports, or at laboratories approved by the Executive Officer of the Regional Water Board.
3. Whenever the Discharger monitors any pollutant at a specified monitoring location more frequently than is required by this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the discharge monitoring report specified in this Order.
4. For every item of monitoring data where the requirements are not met, the monitoring report shall include a statement discussing the reasons for noncompliance, the actions undertaken or proposed that will bring the discharge into full compliance with requirements at the earliest time, and an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Regional Water Board by letter when compliance with the time schedule has been achieved.
5. The Discharger shall assure that records of all monitoring information are maintained and accessible for a period of at least five years from the date of the sample, report, or application. This period of retention shall be extended during the course of any unresolved litigation regarding this discharge or by the request of the Regional Water Board at any time. Records of monitoring information shall include:
 - a. The laboratory which performed the analyses;
 - b. The date(s) analyses were performed;
 - c. The individual(s) who performed the analyses;
 - d. The modification(s) to analytical techniques or methods used;
 - e. All sampling and analytical results, including
 - (1) Units of measurement;

- (2) Minimum reporting level (RL) for the analysis or the minimum level (ML);
 - (3) Results less than the reporting level (RL) or the minimum level (ML), but above the method detection limit (MDL);
 - (4) Data qualifiers and a description of the qualifiers;
 - (5) Quality control test results (and a written copy of the laboratory quality assurance plan);
 - (6) Dilution factors, if used; and
 - (7) Sample matrix type.
 - f. All monitoring equipment calibration and maintenance records;
 - g. All original strip charts from continuous monitoring devices;
 - h. All data used to complete the application for this Order;
 - i. Copies of all reports required by this Order; and,
 - j. Electronic data and information generated by the Supervisory Control and Data Acquisition (SCADA) System.
6. The flow measurement system shall be calibrated at least once per year or more frequently, to ensure continued accuracy.
7. All monitoring instruments and devices used by the Discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy. In the event that continuous monitoring equipment is out of service for greater than a 24-hour period, the Discharger shall obtain a representative grab sample each day the equipment is out of service. The Discharger shall correct the cause(s) of failure of the continuous monitoring equipment as soon as practicable. In its monitoring report, the Discharger shall specify the period(s) during which the equipment was out of service and if the problem has not been corrected, shall identify the steps which the Discharger is taking or proposes to take to bring the equipment back into service and the schedule for these actions.
8. Monitoring and reporting shall be in accordance with the following:
- a. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity.
 - b. The monitoring and reporting of influent, effluent, and sludge shall be done more frequently, as necessary, to maintain compliance with this Order and/or as specified in this Order.
 - c. A "grab" sample is defined as any individual sample collected in less than 15 minutes.

- d. A composite sample is defined as a combination of no fewer than eight individual grab samples obtained over the specified sampling period. The volume of each individual grab sample shall be proportional to the discharge flow rate at the time of sampling. The compositing period shall equal the specific sampling period, or 24 hours, if no period is specified.
- e. Daily samples shall be collected on each day of the week.
- f. Weekly samples shall be collected on any representative day during a week.
- g. Monthly samples shall be collected on any representative day of each month.

II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order.

Table 1. Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description	Latitude	Longitude
(Influent)	M-INF1	WRP influent at the headworks	33°37'05"N	117°43'31"W
DP-001	REC-001	WRP Recycled Water Effluent	33°37'08"N	117°43'37"W

III. INFLUENT MONITORING REQUIREMENTS

A. Influent Monitoring

1. The Discharger shall monitor the influent to the Water Recycling Plant at the influent monitoring location specified in Table 1, above, for the following constituents.

Table 2. Influent Monitoring at M-INF1

Parameter	Units	Sample Type	Minimum Sampling Frequency
Flow	mgd	Recorder/Totalizer	Continuous
pH	pH Units	Recorder	Continuous
Specific Conductance	µmhos/cm	Recorder	Continuous
BOD ₅	mg/L	Composite	Weekly
Total Suspended Solids	mg/L	Composite	Weekly

Table 2. Influent Monitoring at M-INF1

Parameter	Units	Sample Type	Minimum Sampling Frequency
Total Dissolved Solids (TDS)	mg/L	Composite	Quarterly
Total Inorganic Nitrogen (TIN)	mg/L	Composite	Quarterly
Ammonia-Nitrogen	mg/L	Composite	Quarterly

IV. RECLAMATION MONITORING REQUIREMENTS

A. Recycled Water Monitoring Location REC-001

1. The Discharger shall monitor the recycled water entering its distribution system at monitoring location REC-001, as specified in Table 1, above, for the following constituents.

Table 3. Recycled Water Monitoring at REC-001

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency
Flow	mgd	Recorder/Totalizer	Continuous
pH	Standard units	Recorder	Continuous
CT ¹	mg/L-min	Calculation	Continuous
Turbidity ²	NTU	Recorder	Continuous
Coliform Organisms	MPN per 100 mL	Grab	Daily
BOD ₅	mg/L	composite	Weekly
Total Suspended Solids	mg/L	composite	Weekly
TDS	mg/L	composite	Quarterly
TIN	mg/L	Composite	Quarterly

¹ CT is the product of total chlorine residual and modal contact time measured at the same point.

² Turbidity analysis shall be continuous, performed by a continuous recording turbidity analyzer. The results of the daily average turbidity determinations shall be reported monthly. Turbidity samples shall be collected after filtration units but before disinfection.

Table 3. Recycled Water Monitoring at REC-001

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency
Volatile organic portion of EPA Priority Pollutants (See Attachment "G") ³	µg/L	Grab	Annually (during the month of August)
Remaining EPA Priority Pollutants (See Attachment "G") ⁴	µg/L	Composite	Annually (during the month of August)

B. Monitoring Recycled Water Users

Whenever recycled water is supplied to a user, the Discharger shall record on a permanent log: the volume of recycled water supplied; the user of recycled water; the locations of those sites; the type of use (e.g. irrigation, industrial, etc); and the dates on which water is supplied.

The Discharger shall submit a quarterly report summarizing reclaimed water use, including the total amount of reclaimed water supplied, the total number of reclaimed water use sites, and the locations of those sites.

VI. OTHER MONITORING REQUIREMENTS

A. Biosolids Monitoring

1. The Discharger shall maintain a permanent log of solids hauled away from the treatment facilities for use/disposal elsewhere, including the date hauled, the volume or weight (in dry tons), type (screening, grit, raw sludge, biosolids), application (agricultural, composting, etc.), and destination.

B. Water Supply Monitoring

1. In August of each year, a sample of each source of the water supplied to the sewered area shall be obtained and analyzed for total dissolved solids concentration expressed in "mg/L".
2. Quarterly reports shall be submitted stating the amount (in percentage or acre-feet) supplied to the sewered area from each source of water and the resulting flow-weighted water supply quality for total dissolved solids.

³ Minimum levels for priority pollutant monitoring are included in Attachment H.

⁴ See Attachment H for minimum levels applicable to priority pollutant monitoring.

C. Pretreatment Monitoring and Reporting – Not applicable

VII. REPORTING REQUIREMENTS

A. General Monitoring and Reporting Requirements

1. All analytical data shall be reported with method detection limit⁵ (MDLs) and with identification of either reporting level (RL), minimum level (ML), or limits of quantitation (LOQs).
2. Laboratory data for effluent samples must quantify each constituent down to the approved reporting levels (RL) or minimum level (ML) for specific constituents. Any internal quality control data associated with the sample must be reported when requested by the Executive Officer. The Regional Water Board will reject the quantified laboratory data if quality control data is unavailable or unacceptable.
3. Discharge monitoring data shall be submitted in a format acceptable by the Regional Water Board. Specific reporting format may include preprinted forms and/or electronic media. The results of all monitoring required by this Order shall be reported to the Regional Water Board, and shall be submitted in such a format as to allow direct comparison with the limitations and requirements of this Order.
4. The Discharger shall tabulate the monitoring data to clearly illustrate compliance and/or noncompliance with the requirements of the Order.
5. For every item of monitoring data where the requirements are not met, the monitoring report shall include a statement discussing the reasons for noncompliance, and the actions undertaken or proposed which will bring the discharge into full compliance with requirements at the earliest time, and an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Regional Water Board by letter when compliance with the time schedule has been achieved.
6. Self Monitoring Reports (SMRs) shall be signed by either the principal Executive Officer or ranking elected official. A duly authorized representative of the aforementioned signatories may sign documents if:
 - a. The authorization is made in writing by the signatory;
 - b. The authorization specifies the representative as either an individual or position having responsibility for the overall operation of the regulated facility or activity:
and

⁵

The standardized test procedure to be used to determine the method detection limit (MDL) is given at Appendix B, 'Definition and Procedure for the Determination of the Method Detection Limit' of 40 CFR 136.

- c. The written authorization is submitted to the Executive Officer of the Regional Water Board.
7. SMRs shall contain the following completed declaration:
- “I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.”
8. The monthly reports for June and December shall include a roster of plant personnel, including job titles, duties, and level of State certification for each individual.
9. The Discharger shall prepare a monthly summary of operating records. The monthly summary shall include operational problems, plant and equipment breakdowns, diversion to emergency storage or disposal, and all corrective or preventive action taken. The monthly summary of operating records shall be reported monthly.
10. The Discharger shall report monitoring results for specific parameters in accordance with the following table:

Table 4. Reporting Requirements

Parameter	Measurement
Flow	Daily total flow
pH	Daily High and daily low
Turbidity	Daily maximum and daily average
CT	Daily minimum and daily average

B. Self Monitoring Reports (SMRs)

1. Monitoring Reports shall be submitted by the last day of each month and shall include all data collected during the previous month.
2. At any time during the term of this permit, the State or Regional Water Board may notify the Discharger to electronically submit Self-Monitoring Reports (SMRs) using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site: <http://www.waterboards.ca.gov/ciwqs/index.html>. Until such notification is given, the Discharger shall submit hard copy SMRs. The CIWQS Web site will provide additional directions for SMR submittal in the event there will be service interruption for electronic submittal.
3. Reporting Protocols: The Discharger shall report with each sample result the applicable Reporting Level (RL), or Minimum Level (ML), and the current Method Detection Limit (MDL), as determined by the procedure in 40 CFR Part 136.

The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- a. Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- b. Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The *estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ as well as the words "Estimated Concentration" (may be shortened to "Est. Conc."). The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ($\pm$ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- c. Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
 - d. Dischargers are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from *extrapolation* beyond the lowest point of the calibration curve.
4. The Discharger shall submit hard copy SMRs (with an original signature) when required by subsection B.1, above, in accordance with the following requirements:
- a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations.
 - b. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
 - c. SMRs must be submitted to the Regional Water Board, signed and certified as specified by Reporting Requirement VII.A.6. and VII.A.7. above, to the address listed below:

California Regional Water Quality Control Board
Santa Ana Region
3737 Main Street, Suite 500
Riverside, CA 92501-3348

5. By April 1 of each year, the Discharger shall submit an annual report to the Regional Water Board. The annual report shall include the following:
 - a. Tabular and graphical summaries of the monitoring data obtained during the previous year, including monitoring data for priority pollutants;
 - b. A discussion of the compliance record and the corrective actions taken or planned, which may be needed to bring the discharge into full compliance with the waste discharge requirements; and
 - c. A summary of the quality assurance (QA) activities for the previous year.

Attachment F – FACT SHEET

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ATTACHMENT F – FACT SHEET

As described in Section II of this Order, this Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Dischargers in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Discharger.

I. DISCHARGER/FACILITY INFORMATION

The following table summarizes administrative information related to the Discharger/Facility:

Table 1. Discharger/Facility Information	
WDID	8 303598001
Discharger/Operator	El Toro Water District
Name of Facility	Water Recycling Plant
Facility Address	23542 Moulton Parkway, Laguna Woods, CA 92654
Facility Contact, Title and Phone	Mark Pade, Chief Plant Operator, (949) 837-7050, Ext 103, mpade@etwd.com
Authorized Person to Sign and Submit Reports	Dennis Cafferty, Director of Operations and Engineering, (949) 837-7050, Ext 223, dcafferty@etwd.com
Mailing/Billing Address	24251 Los Alisos Boulevard, Lake Forest, CA 92630
Type of Facility	POTW
Threat to Water Quality	2
Complexity	B
Pretreatment Program	N
Reclamation Requirements	Producer/User
Facility Design Flow – secondary treatment	6 million-gallons-per-day (mgd) secondary treated
Facility Design Flow – tertiary treatment	4.0 mgd tertiary treated and disinfected
Facility Permitted Flow	6.0 mgd
Receiving Water	Irvine Groundwater Management Zone
Receiving Water Type	Groundwater
Watershed	San Diego Creek/Upper Newport Bay Watershed

- A. El Toro Water District (hereinafter Discharger or ETWD) owns and operates a sanitary sewer collection system, a Water Recycling Plant (hereinafter the Facility), and a recycled water distribution system. The Discharger currently discharges tertiary treated disinfected wastewater from the Facility pursuant to waste discharge and producer/user water reclamation requirements, Order No. 94-3. The Santa Ana Regional Water Board issued Order No. 94-3 to the South Orange County Reclamation Authority (SOCRA, predecessor to South Orange County Wastewater Authority or SOCWA). ETWD is a member of SOCWA. SOCWA was created to facilitate and manage the collection, transmission, treatment and disposal of wastewater from South Orange County areas. Other member agencies of SOCWA include: Irvine Ranch Water District, City of Laguna Beach, Moulton Niguel Water District, City of San Clemente, South Coast Water District, City of San Juan Capistrano, Santa Margarita Water District, Trabuco Canyon Water District and Emerald Bay Service District. Most facilities under SOCWA are within the San Diego Regional Water Board's jurisdiction. ETWD facilities are mostly within the Santa Ana Regional Water Board's jurisdiction. As indicated in Table 1, the Facility has secondary treatment capacity for 6 mgd and tertiary treatment capacity for 4 mgd. Secondary treated un-disinfected wastewater is discharged to the Pacific Ocean via the Aliso Creek ocean outfall and it is regulated under San Diego Regional Water Board Order No. R9-2012-0013, NPDES Permit No. CA0107611, issued to SOCWA. This Order only regulates the discharge of tertiary treated disinfected recycled water within the Santa Ana Regional Board's jurisdiction.
- B. The Discharger submitted a Title 22 Engineering Report (Title 22 Report) to the California Department Public Health (CDPH, now the State Water Resources Control Board, Division of Drinking Water) on February 5, 2013, pursuant to section 60323, Article 7, Chapter 3, Division 4, Title 22 of the California Code of Regulations. The Title 22 Report contained pertinent technical details regarding the design and operation of ETWD's Water Recycling Plant and the proposed tertiary treatment and recycled water distribution systems. CDPH reviewed the Title 22 Report and provided comments to ETWD through an email message dated March 6, 2013. On June 12, 2014, ETWD submitted an updated Report dated June 2014 and other requested items to CDPH for its review and approval. CDPH staff responded with an email message dated June 25, 2014, in which CDPH staff indicated that they had no further comments regarding the Title 22 Report. The facility was inspected by Regional Water Board staff on December 18, 2014.
- C. On June 17, 2009, SOCWA filed a report of waste discharge and submitted an application for renewal of Order No. 94-3. SOCWA's application was triggered by several events, which included the following: 1) the South Orange County Reclamation Authority changed its name to the South Orange County Wastewater Authority; 2) Los Alisos Water District became part of Irvine Ranch Water District (IRWD); 3) Los Alisos Water Recycling Plant (LAWRP) expanded its recycled water production capacity to 7.1 mgd; and 4) IRWD built an interconnection between its Michelson Water Recycling Plant's (MWRP's) recycled water distribution system and LAWRP's recycled water distribution system.

- D. Based on the acquisition of Los Alisos Water District and its wastewater treatment plant (LAWRP) by IRWD, the interconnection of the recycled water systems between MWRP and LAWRP, and the tertiary treatment system upgrade built by El Toro Water District, Regional Water Board staff decided not to renew WDRs for SOCWA. IRWD agreed to consolidate its water recycling plants under one permit and ETWD agreed to have a separate waste discharge requirements for its discharges within the Santa Ana Regional Water Board's jurisdiction. Both ETWD and LAWRP may discharge secondary treated wastewater through transmission mains to the Aliso Creek Ocean outfall and the transmission mains are owned by SOCWA. However, SOCWA does not own the IRWD and ETWD's treatment facilities and their recycled water distribution systems.

II. FACILITY DESCRIPTION

A. Description of Wastewater and Biosolids Treatment and/or Controls

1. Facility Background

This Facility is a publicly owned treatment works with preliminary and secondary treatment capacity of 6 million-gallons-per-day (mgd), tertiary treatment and disinfection capacity of 4 mgd. In the past, it has produced up to 1.2 mgd of disinfected secondary treated recycled water used for mainly landscape irrigation. The Discharger has recently completed improvements to the treatment capabilities of the Facility by constructing a new tertiary treatment and recycled water distribution systems that consist of two rotating-disk cloth-media filters, tertiary effluent transfer pumps, a recycled water storage tank with integrated chlorine contact basin (dual purpose), a recycled water pump station, an alum/poly blend storage and feed station, and approximately 19 miles of pipelines for distribution of recycled water, among other improvements. As of February 2015, the Discharger has started the distribution of tertiary treated and disinfected recycled water to new use areas connected to its distribution pipeline and expects to distribute up to 4.0 mgd of recycled water in the future as new use areas are connected, tested and approved.

2. Service Area

This Facility treats residential and commercial wastes from portions of the Cities of Lake Forest, Laguna Hills, Aliso Viejo, Mission Viejo, and all of the City of Laguna Woods. The total population served is over 51,000. The collection system is composed of 142 miles of sewer, 11 lift stations and 1 large gravity zone. Raw sewage arrives at the Facility through a 24-inch gravity main and a 16-inch force main that merge at a manhole and into an influent trunkline prior to the headworks.

3. Design Characteristics

The following table summarizes the treatment processes at the Water Recycling Plant:

Table 2. Plant Design

Preliminary Treatment	Influent flow metering, mechanical bar-screen, aerated grit chamber, fine screening (roto-strainers), influent flow equalization, and odor control system.
Primary Treatment	No primary treatment
Secondary Treatment	6.0 mgd conventional activated sludge process 2.5-acre holding pond (used to temporarily store secondary effluent when disposal through the ocean outfall is not available or limited during emergency conditions)
Tertiary Treatment	4.0 mgd Rotating-disk cloth filtration, chlorine disinfection
Solids Handling	Waste activated sludge aeration coupled with dewatering with diffused air flotation thickening (DAFT). Screenings from headworks and sludge are hauled away to the Aliso Water Management Agency's joint facilities for composting.

4. Recycled Water Use

In the past, most of the disinfected secondary treated recycled water produced by the Discharger was used for irrigation at the Laguna Woods Village Golf Course. Also, some of the recycled water is used within the Facility for plant utility water. This recycled water distribution system included one pump station and less than 1 mile of pipeline. The completion of the tertiary treatment and disinfection upgrade has added one recycled water storage tank with integrated chlorine contact basin (with an approximate capacity of 2 million gallons), a recycled water pump station, and about 19 miles of recycled water distribution pipeline to reach new use areas. The Facility was able to supply approximately 500 acre-feet per year of disinfected secondary treated recycled water to the Laguna Woods Village Golf Course. Currently, with the tertiary treatment and disinfection upgrade and new recycled water distribution system in operation, ETWD envisions producing up to 1,400 acre-feet per year of tertiary-2.2 recycled water for its use areas and has discontinued the production and distribution of secondary treated disinfected recycled water. The production of tertiary-2.2 recycled water allows the Discharger to use the recycled water for all categories of reuse, except for indirect and direct potable water reuse.

B. Discharge Points and Receiving Waters

Following table shows the discharge point, longitude and latitude, affected receiving waters, and estimated volume of discharge:

Table 3. Discharge Points

Discharge Serial No.	Effluent Description	Location (Latitude & Longitude)	Receiving Water	Flow & Frequency
DP-001 Recycled Water	Disinfected Tertiary	33°37'08"N 117°43'37"W	Irvine Groundwater Management Zone	Up to 4.0 mgd, intermittent

1. Discharge Point for Recycled Water Use – DP-001

The Discharger has started the distribution of tertiary-2.2 recycled water and plans to distribute up to 3.7 mgd of recycled water via DP-001 to the ETWD's recycled water system. The final design capacity of the tertiary treatment and disinfection system is 4.0 mgd. The recycled water use area overlies the Irvine Groundwater Management Zone.

2. Stormwater

Industrial Stormwater Requirements: Pursuant to Section 402(p) of the Clean Water Act and Title 40 of the Code of Federal Regulations (CFR) Part 122, 123, and 124, the State Water Resources Control Board adopted a general NPDES permit to regulate storm water discharges associated with industrial activities (State Board Order No. 97-03-DWQ) on April 17, 1997. Storm water discharges from the Facility are regulated under the State Board Order No. 97-03-DWQ. As of July 1, 2015, Order No. 97-03-DWQ will be replaced by Order No. 2014-0057-DWQ.

III. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in the proposed Order are based on the requirements and authorities described in this section.

A. Legal Authorities

This Order serves as Waste Discharge Requirements pursuant to article 4, Chapter 4, Division 7 of the Water Code (commencing with Section 13260). Also, this Order serves as a Master Reclamation Permit pursuant to Section 13523.1 of Article 4, Chapter 7, Division 7 of the California Water Code.

B. California Environmental Quality Act (CEQA)

This action also involves the re-issuance of waste discharge requirements for an existing facility that discharges treated wastewater to land and as such, is exempt from the provisions of California Environmental Quality Act (commencing with Section 21100) in that the activity is exempt pursuant to Title 14 of the California Code of Regulations Section 15301. In compliance with the California Environmental Quality Act (CEQA), the Discharger completed a separate CEQA study for the tertiary treatment upgrade of the Facility and for the recycled water distribution system expansion. A Mitigated Negative Declaration was certified by El Toro Water District and a Notice of Determination was filed for both projects on March 30, 2012 (SCH# 2012031096 and 2012031100, respectively). The Initial Study/Mitigated Negative Declaration for each separate project concluded that both projects would have no impact or less-than-significant impact provided certain mitigation measures were taken during the construction phase. The Discharger also obtained authorization from the State Water Resources Control Board's Division of Drinking Water for the production and distribution of tertiary-2.2 recycled water. Regional Water Board staff reviewed the environmental documents prepared for the Facility and recycled water distribution and determined that if the requirements specified herein are complied with, there should not be any significant water quality or other adverse environmental impacts from the permitted discharges.

C. State Regulations, Policies, and Plans

1. Water Quality Control Plans

The Regional Water Board adopted a Water Quality Control Plan for the Santa Ana River Basin (hereinafter Basin Plan) that became effective on January 24, 1995. The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the Basin Plan.

On January 22, 2004, the Regional Water Board adopted Resolution No. R8-2004-0001, amending the Basin Plan to incorporate revised boundaries for groundwater subbasins, now termed "management zones", new nitrate-nitrogen and TDS objectives for the new management zones, and new nitrogen and TDS management strategies applicable to both surface and ground waters. The State Water Board and Office of Administrative Law (OAL) approved the N/TDS Amendment on September 30, 2004 and December 23, 2004, respectively. Accordingly, these waste discharge requirements implement relevant, groundwater-related components of the N/TDS Amendment. Specifically, the total dissolved solids (TDS) limitation established in this Order is based on the amended Basin Plan.

As previously discussed, the Facility is producing tertiary treated recycled water that is being used in areas overlying the Irvine Groundwater Management Zone. The designated beneficial uses of receiving waters affected by the discharge from the Facility are as follows:

Table 4. Basin Plan Beneficial Uses

Discharge Point	Receiving Water	Beneficial Uses
001	Irvine GMZ	<u>Present or Potential:</u> a. Municipal and domestic supply, b. Agricultural supply, c. Industrial service supply, and d. Industrial process supply

Requirements of this Order implement the Basin Plan.

2. Antidegradation Policy

The State Water Board established California's antidegradation policy in State Water Board Resolution No. 68-16. Resolution No. 68-16 incorporates the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Regional Water Board's Basin Plan implements, and incorporates by reference the State antidegradation policies. Requirements specified in this Order should prevent any degradation of the receiving waters. Therefore, the permitted discharge is consistent with the antidegradation provisions of State Water Board Resolution No. 68-16.

3. Pretreatment

Although the design capacity of the Water Recycling Plant is more than 5 mgd (6 mgd actual capacity), there are no significant industrial users within the service area and no episodes of interference, plant upset, and/or pass-through have occurred. Consequently, this Order does not contain requirements for a pretreatment program pursuant to Sections 2233 and 2235.3, Title 23, of the California Code of Regulations. (See Section VI.C.5.c., below)

4. Biosolids Requirements

On February 19, 1993, the USEPA issued a final rule for the use and disposal of sewage sludge, 40 CFR, Part 503. This rule requires that producers of sewage sludge meet certain reporting, handling, and disposal requirements. The State of California has not been delegated the authority to implement this program, therefore, the U.S. Environmental Protection Agency is the implementing agency. The sludge generated by the Facility is hauled away to the Aliso Water Management Agency's joint facilities for composting; therefore, this Order does not include biosolids requirements.

5. Monitoring and Reporting Requirements

Section 13267 of the CWC authorizes the Regional Water Board to require technical and monitoring reports. The Monitoring and the Reporting Program (MRP) establishes monitoring and reporting requirements to implement state requirements. This MRP is provided in Attachment E.

D. Other Plans, Policies and Regulations – Not Applicable

IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

A. Discharge Prohibitions

The discharge prohibitions are based on the Basin Plan and State Water Resources Control Board's plans and policies. These prohibitions are consistent with the requirements set for other discharges regulated by waste discharge requirements adopted by the Regional Water Board.

B. Technology-Based Effluent Limitations – Not Applicable

C. Water Quality-Based Effluent Limitations (WQBELs) For Land Disposal – Not Applicable

D. Best Professional Judgment-Based Effluent Limitations

For tertiary treated wastewater, the BOD₅ and TSS concentration limits shown in the following table are based on Best Professional Judgment.

Table 5. Effluent BOD₅ and TSS Limits for Tertiary Effluent

Constituent	Average Weekly	Average Monthly
Biochemical Oxygen Demand	30 mg/L	20 mg/L
Suspended Solids	30 mg/L	20 mg/L

E. Summary of Land Disposal Limitations – Not Applicable

F. Reclamation Specifications: DP-001

1. Section 13523 of the California Water Code provides that a Regional Water Board, after consulting with and receiving the recommendations from the California Department of Public Health (CDPH) [after July 1, 2014, California State Water

Resources Control Board's (SWRCB's) Division of Drinking Water (DDW)] and any party who has requested in writing to be consulted, and after any necessary hearing, shall prescribe water reclamation requirements for water which is used or proposed to be used as recycled water, if, in the judgment of the Regional Water Board, such requirements are necessary to protect the public health, safety, or welfare. Section 13523 further provides that such requirements shall include, or be in conformance with, the statewide uniform water recycling criteria established by CDPH (or the California SWRCB's DDW) pursuant to California Water Code Section 13521.

2. Reclamation specifications in the proposed Order are based on the recycling criteria contained in Title 22, Division 4, Chapter 3, Sections 60301 through 60355, California Code of Regulations, and California Water Code Section 13521.
3. For recycled water use, this Order (Section IV.A.2. Reclamation Specifications – Discharge Point 001) specifies TDS limits based on the water quality objective for the Irvine Groundwater Management Zone, which is 910 mg/l.
4. Summary of Recycled Water Limitations:

Table 6. Summary of Recycled Water Limitations at DP-001

Parameter	Units	Effluent Limitations					Basis
		Average Monthly or as noted herein	Average Weekly	Daily average or as noted herein	Instantaneous Minimum or as noted herein	Instantaneous Maximum	
BOD ₅	mg/L	20	30	--	--	--	BPJ
Total Suspended Solids	mg/L	20	30	--	--	--	BPJ
CT	milligram-minutes per liter	--	--	--	450	--	Title 22
Coliform	MPN/100 mL	23 ¹	2.2 median in 7 days	240 ²	--	--	Title 22
Turbidity	NTU	--	--	2	5 ³	10	Title 22

5. The Basin Plan water quality objectives for TIN do not apply to recycled water use for agricultural and landscape irrigation because of plant uptake of nitrogen.

¹ Not more than once in any 30-day period

² Maximum Daily

³ Not to exceed more than 72 minutes in any 24-hour period

V. RATIONALE FOR RECEIVING WATER LIMITATIONS

A. Surface Water – Not Applicable

B. Groundwater

1. The receiving groundwater limitations in the proposed Order are based on the water quality objectives contained in the Basin Plan.

VI. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

Section 13267 of the CWC authorizes the Water Boards to require technical and monitoring reports. The Monitoring and Reporting Program (MRP), Attachment E of this Order, establishes monitoring and reporting requirements to implement federal and State requirements. The following provides the rationale for monitoring and reporting requirements contained in the MRP.

A. Influent Monitoring

Influent monitoring is required to evaluate the need for a pretreatment program and to assess treatment plant performance.

B. Effluent Monitoring

The Discharger is required to conduct monitoring of the permitted discharges in order to evaluate compliance with permit conditions. Monitoring requirements are included in the proposed MRP. The MRP includes Standard Monitoring Provisions. These provisions are standard requirements applicable to all waste discharge requirements issued by the Regional Water Board. In addition to containing definitions of terms, they specify general sampling/analytical protocols and requirements for reporting of spills, violations, and routine monitoring data in accordance with California Code of Regulations, the California Water Code, and Regional Water Board's policies. The MRP also contains specific sampling requirements for the Discharger's wastewater treatment plant. It defines the sampling stations, frequency, pollutants to be monitored, and additional reporting requirements. Pollutants to be monitored include all pollutants for which effluent limitations are specified. Effluent limits are included for all pollutants for which there are a reasonable potential to be present in the discharge.

C. Other Monitoring Requirements

1. Water Supply Monitoring

The Discharger is required to collect a sample of each source of water supplied and analyze for total dissolved solids. The result of this monitoring will enable the Discharger to show compliance with TDS limitations in the Order.

2. Biosolids Monitoring

Biosolids produced in this facility are sent to the Aliso Water Management Agency's joint facilities for composting. As such, requirements for biosolids monitoring are not included in this Order.

3. Pretreatment Monitoring

Although the design capacity of the Facility is greater than 5 mgd, there are no significant industrial users within the service area and the Facility has not experienced episodes of upset, pass-through, and/or interference. Consequently, this Order does not contain requirements for the implementation of an effective pretreatment program pursuant to Sections 2233 and 2235.3, Title 23, of the California Code of Regulations.

VII. RATIONALE FOR PROVISIONS

A. Provisions

1. Special Studies and Additional Monitoring Requirements

This Order requires the Discharger to submit a TDS mitigation plan to mitigate any discharge that exceeds the TDS limits in the Order. Since priority pollutants are not expected in the discharge, priority pollutant monitoring is only required on an annual basis.

2. Construction, Operation, and Maintenance Specifications

The Facility produces and distributes tertiary treated disinfected recycled water for various uses and, therefore, must implement proper staffing, operation, maintenance, design, and reliability requirements contained in the Water Recycling Criteria (Title 22, Division 4, Chapter 3, Articles 7 through 10, California Code of Regulations). Also, the Discharger must ensure that the operations staff has the proper certification as required under Title 23, Division 3, Chapter 26, California Code of Regulations.

3. Special Provisions for Municipal Facility - POTWs Only

a. Sewer Collection System Requirements

1. The State Water Board issued the Statewide General Waste Discharge Requirements for Sanitary Sewer Systems, Order No. 2006-0003-DWQ (General Order) on May 2, 2006, as amended by Order No. WQ. 2008-0002-EXEC. The General Order requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order requires

agencies to develop sanitary sewer management plans and report all sanitary sewer overflows, among other requirements and prohibitions. The Discharger has obtained coverage under the General Order and implemented these requirements.

Furthermore, the General Order contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. Inasmuch as the Discharger's collection system is part of the system that is subject to this Order, Provision section VI.C.4.a. of this Order applies. For instance, the 24-hour reporting requirements in this Order (Provision section VI.A.2.) are not included in the General Order. The Discharger must comply with both the General Order and this Order.

b. Sludge and Other Waste Disposal Requirements

1. Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with State Water Board and Integrated Waste Management Board's joint regulations (Title 27) of the California Code of Regulations and approved by the Regional Water Board's Executive Officer.

c. Pretreatment Program Requirements

1. Although the design capacity of the Facility is greater than 5 mgd, there are no significant industrial users within the service area and the Facility has not experienced episodes of upset, pass-through, and/or interference. Consequently, this Order does not contain requirements for the implementation of an effective pretreatment program pursuant to Sections 2233 and 2235.3, Title 23, of the California Code of Regulations.

VIII. PUBLIC PARTICIPATION

The Regional Water Board will be considering the adoption of waste discharge requirements for El Toro Water District's Water Recycling Plant. As a step in the waste discharge requirements adoption process, the Regional Water Board staff has developed tentative waste discharge requirements. Regional Water Board staff circulated the draft Order to interested parties and made it available for public review and comments on its website.

A. Notification of Interested Parties

The Regional Water Board staff has notified the Discharger and interested agencies and persons of the Board's intent to prescribe waste discharge requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Notification was provided through mailings and at the Regional Water Board website: <http://www.waterboards.ca.gov/santaana>.

B. Written Comments

The staff determinations are tentative. Interested persons are invited to submit written comments concerning this tentative Order. To be fully responded to by staff and considered by the Regional Water Board, written comments should be received at the Regional Water Board offices at the address given below by 5:00 p.m. on June 2, 2015.

Julio Lara (julio.lara@waterboards.ca.gov)
California Regional Water Quality Control Board
Santa Ana Region
3737 Main Street, Suite 500
Riverside, CA 92501-3348

C. Public Hearing

The Regional Water Board will hold a public hearing on the tentative Order during its regular Board meeting on the following date and time and at the following location:

Date: June 19, 2015
Time: 9:00 A.M.
Location: City Council Chambers
City of Loma Linda
25541 Barton Road
Loma Linda, CA 92354

Interested persons are invited to attend. At the public hearing, the Regional Water Board will hear testimony, if any, pertinent to the discharge and the tentative Order. Oral testimony will be heard; however, for accuracy of the record, important testimony should be in writing.

Please be aware that dates and venues may change. Our web address, where you can access the current agenda for changes in dates and locations is:
<http://www.waterboards.ca.gov/santaana>.

D. Waste Discharge Requirements Petitions

Any aggrieved person may petition the State Water Board to review the decision of the Regional Water Board regarding the final Order. The petition must be submitted within 30 days of the Regional Water Board's action to the following address:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100
Sacramento, CA 95812-0100

E. Information and Copying

The Report of Waste Discharge, related documents, tentative order, comments received, and other information are on file and may be inspected at the address above at any time between 9:00 a.m. and 3:00 p.m. Monday through Friday. Copying of documents may be arranged through the Regional Water Board by calling (951) 782-4901 or Julio.lara@waterboards.ca.gov.

F. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the Waste Discharge Requirements, NPDES permits, or any other matter that the Regional Water Board considers should register at:

http://www.waterboards.ca.gov/resources/email_subscriptions/reg8_subscribe.shtml

G. Additional Information

Requests for additional information or questions regarding this Order should be directed to Julio Lara at (951) 782-4901 or Julio.lara@waterboards.ca.gov.

ATTACHMENT G - EPA PRIORITY POLLUTANT LIST

EPA PRIORITY POLLUTANT LIST		
Metals	Acid Extractibles	Base/Neutral Extractibles (continuation)
1. Antimony	45. 2-Chlorophenol	91. Hexachloroethane
2. Arsenic	46. 2,4-Dichlorophenol	92. Indeno (1,2,3-cd) Pyrene
3. Beryllium	47. 2,4-Dimethylphenol	93. Isophorone
4. Cadmium	48. 2-Methyl-4,6-Dinitrophenol	94. Naphthalene
5a. Chromium (III)	49. 2,4-Dinitrophenol	95. Nitrobenzene
5b. Chromium (VI)	50. 2-Nitrophenol	96. N-Nitrosodimethylamine
6. Copper	51. 4-Nitrophenol	97. N-Nitrosodi-N-Propylamine
7. Lead	52. 3-Methyl-4-Chlorophenol	98. N-Nitrosodiphenylamine
8. Mercury	53. Pentachlorophenol	99. Phenanthrene
9. Nickel	54. Phenol	100. Pyrene
10. Selenium	55. 2, 4, 6 – Trichlorophenol	101. 1,2,4-Trichlorobenzene
11. Silver	Base/Neutral Extractibles	Pesticides
12. Thallium	56. Acenaphthene	102. Aldrin
13. Zinc	57. Acenaphthylene	103. Alpha BHC
Miscellaneous	58. Anthracene	104. Beta BHC
14. Cyanide	59. Benzidine	105. Delta BHC
15. Asbestos (not required unless requested)	60. Benzo (a) Anthracene	106. Gamma BHC
16. 2,3,7,8-Tetrachlorodibenzo-P-Dioxin (TCDD)	61. Benzo (a) Pyrene	107. Chlordane
Volatile Organics	62. Benzo (b) Fluoranthene	108. 4, 4' - DDT
17. Acrolein	63. Benzo (g,h,i) Perylene	109. 4, 4' - DDE
18. Acrylonitrile	64. Benzo (k) Fluoranthene	110. 4, 4' - DDD
19. Benzene	65. Bis (2-Chloroethoxy) Methane	111. Dieldrin
20. Bromoform	66. Bis (2-Chloroethyl) Ether	112. Alpha Endosulfan
21. Carbon Tetrachloride	67. Bis (2-Chloroisopropyl) Ether	113. Beta Endosulfan
22. Chlorobenzene	68. Bis (2-Ethylhexyl) Phthalate	114. Endosulfan Sulfate
23. Chlorodibromomethane	69. 4-Bromophenyl Phenyl Ether	115. Endrin
24. Chloroethane	70. Butylbenzyl Phthalate	116. Endrin Aldehyde
25. 2-Chloroethyl Vinyl Ether	71. 2-Chloronaphthalene	117. Heptachlor
26. Chloroform	72. 4-Chlorophenyl Phenyl Ether	118. Heptachlor Epoxide
27. Dichlorobromomethane	73. Chrysene	119. PCB 1016
28. 1,1-Dichloroethane	74. Dibenzo (a,h) Anthracene	120. PCB 1221
29. 1,2-Dichloroethane	75. 1,2-Dichlorobenzene	121. PCB 1232
30. 1,1-Dichloroethylene	76. 1,3-Dichlorobenzene	122. PCB 1242
31. 1,2-Dichloropropane	77. 1,4-Dichlorobenzene	123. PCB 1248
32. 1,3-Dichloropropylene	78. 3,3'-Dichlorobenzidine	124. PCB 1254
33. Ethylbenzene	79. Diethyl Phthalate	125. PCB 1260
34. Methyl Bromide	80. Dimethyl Phthalate	126. Toxaphene
35. Methyl Chloride	81. Di-n-Butyl Phthalate	
36. Methylene Chloride	82. 2,4-Dinitrotoluene	
37. 1,1,2,2-Tetrachloroethane	83. 2-6-Dinitrotoluene	
38. Tetrachloroethylene	84. Di-n-Octyl Phthalate	
39. Toluene	85. 1,2-Diphenylhydrazine	
40. 1,2-Trans-Dichloroethylene	86. Fluoranthene	
41. 1,1,1-Trichloroethane	87. Fluorene	
42. 1,1,2-Trichloroethane	88. Hexachlorobenzene	
43. Trichloroethylene	89. Hexachlorobutadiene	
44. Vinyl Chloride	90. Hexachlorocyclopentadiene	

ATTACHMENT H – MINIMUM LEVELS

MINIMUM LEVELS IN PPB (µg/l)

Table 1- VOLATILE SUBSTANCES ¹	GC	GCMS
Acrolein	2.0	5
Acrylonitrile	2.0	2
Benzene	0.5	2
Bromoform	0.5	2
Carbon Tetrachloride	0.5	2
Chlorobenzene	0.5	2
Chlorodibromomethane	0.5	2
Chloroethane	0.5	2
Chloroform	0.5	2
Dichlorobromomethane	0.5	2
1,1 Dichloroethane	0.5	1
1,2 Dichloroethane	0.5	2
1,1 Dichloroethylene	0.5	2
1,2 Dichloropropane	0.5	1
1,3 Dichloropropylene (volatile)	0.5	2
Ethylbenzene	0.5	2
Methyl Bromide (<i>Bromomethane</i>)	1.0	2
Methyl Chloride (<i>Chloromethane</i>)	0.5	2
Methylene Chloride (<i>Dichloromethane</i>)	0.5	2
1,1,2,2 Tetrachloroethane	0.5	1
Tetrachloroethylene	0.5	2
Toluene	0.5	2
trans-1,2 Dichloroethylene	0.5	1
1,1,1 Trichloroethane	0.5	2
1,1,2 Trichloroethane	0.5	2
Trichloroethylene	0.5	2
Vinyl Chloride	0.5	2
1,2 Dichlorobenzene (volatile)	0.5	2
1,3 Dichlorobenzene (volatile)	0.5	2
1,4 Dichlorobenzene (volatile)	0.5	2

Selection and Use of Appropriate ML Value:

ML Selection: When there is more than one ML value for a given substance, the discharger may select any one of those ML values, and their associated analytical methods, listed in this Attachment that are below the calculated effluent limitation for compliance determination. If no ML value is below the effluent limitation, then the discharger shall select the lowest ML value, and its associated analytical method, listed in the PQL Table.

ML Usage: The ML value in this Attachment represents the lowest quantifiable concentration in a sample based on the proper application of all method-based analytical procedures and the absence of any matrix interferences. Assuming that all method-specific analytical steps are followed, the ML value will also represent, after the appropriate application of method-specific factors, the lowest standard in the calibration curve for that specific analytical technique. Common analytical practices sometimes require different treatment of the sample relative to calibration standards.

Note: chemical names in parenthesis and italicized is another name for the constituent.

MINIMUM LEVELS IN PPB (µg/l)

Table 2 – Semi-Volatile Substances²	GC	GCMS	LC
2-Chloroethyl vinyl ether	1	1	
2 Chlorophenol	2	5	
2,4 Dichlorophenol	1	5	
2,4 Dimethylphenol	1	2	
4,6 Dinitro-2-methylphenol	10	5	
2,4 Dinitrophenol	5	5	
2- Nitrophenol		10	
4- Nitrophenol	5	10	
4 Chloro-3-methylphenol	5	1	
2,4,6 Trichlorophenol	10	10	
Acenaphthene	1	1	0.5
Acenaphthylene		10	0.2
Anthracene		10	2
Benzidine		5	
Benzo (a) Anthracene (1,2 Benzanthracene)	10	5	
Benzo(a) pyrene (3,4 Benzopyrene)		10	2
Benzo (b) Fluoranthene (3,4 Benzofluoranthene)		10	10
Benzo(g,h,i)perylene		5	0.1
Benzo(k)fluoranthene		10	2
bis 2-(1-Chloroethoxyl) methane		5	
bis(2-chloroethyl) ether	10	1	
bis(2-Chloroisopropyl) ether	10	2	
bis(2-Ethylhexyl) phthalate	10	5	
4-Bromophenyl phenyl ether	10	5	
Butyl benzyl phthalate	10	10	
2-Chloronaphthalene		10	
4-Chlorophenyl phenyl ether		5	
Chrysene		10	5
Dibenzo(a,h)-anthracene		10	0.1
1,2 Dichlorobenzene (semivolatile)	2	2	
1,3 Dichlorobenzene (semivolatile)	2	1	
1,4 Dichlorobenzene (semivolatile)	2	1	
3,3' Dichlorobenzidine		5	
Diethyl phthalate	10	2	
Dimethyl phthalate	10	2	
di-n-Butyl phthalate		10	
2,4 Dinitrotoluene	10	5	
2,6 Dinitrotoluene		5	
di-n-Octyl phthalate		10	
1,2 Diphenylhydrazine		1	
Fluoranthene	10	1	0.05
Fluorene		10	0.1
Hexachloro-cyclopentadiene	5	5	
1,2,4 Trichlorobenzene	1	5	

MINIMUM LEVELS IN PPB (µg/l)

Table 2 - SEMI-VOLATILE SUBSTANCES²	GC	GCMS	LC	COLOR
Pentachlorophenol	1	5		
Phenol ³	1	1		50
Hexachlorobenzene	5	1		
Hexachlorobutadiene	5	1		
Hexachloroethane	5	1		
Indeno(1,2,3,cd)-pyrene		10	0.05	
Isophorone	10	1		
Naphthalene	10	1	0.2	
Nitrobenzene	10	1		
N-Nitroso-dimethyl amine	10	5		
N-Nitroso -di n-propyl amine	10	5		
N-Nitroso diphenyl amine	10	1		
Phenanthrene		5	0.05	
Pyrene		10	0.05	

Table 3- INORGANICS⁴	FAA	GFAA	ICP	ICPMS	SPGFAA	HYDRIDE	CVAA	COLOR	DCP
Antimony	10	5	50	0.5	5	0.5			1000
Arsenic		2	10	2	2	1		20	1000
Beryllium	20	0.5	2	0.5	1				1000
Cadmium	10	0.5	10	0.25	0.5				1000
Chromium (total)	50	2	10	0.5	1				1000
Chromium VI	5							10	
Copper	25	5	10	0.5	2				1000
Lead	20	5	5	0.5	2				10000
Mercury				0.5			0.2		
Nickel	50	5	20	1	5				1000
Selenium		5	10	2	5	1			1000
Silver	10	1	10	0.25	2				1000
Thallium	10	2	10	1	5				1000
Zinc	20		20	1	10				1000
Cyanide								5	

² With the exception of phenol by colorimetric technique, the normal method-specific factor for these substances is 1000, therefore, the lowest standards concentration in the calibration curve is equal to the above ML value for each substance multiplied by 1000.

³ Phenol by colorimetric technique has a factor of 1.

⁴ The normal method-specific factor for these substances is 1, therefore, the lowest standard concentration in the calibration curve is equal to the above ML value for each substance.

MINIMUM LEVELS IN PPB (µg/l)

Table 4- PESTICIDES – PCBs⁵	GC
Aldrin	0.005
alpha-BHC (<i>a</i> -Hexachloro-cyclohexane)	0.01
beta-BHC (<i>b</i> -Hexachloro-cyclohexane)	0.005
Gamma-BHC (<i>Lindane</i> ; <i>g</i> -Hexachloro-cyclohexane)	0.02
Delta-BHC (<i>d</i> -Hexachloro-cyclohexane)	0.005
Chlordane	0.1
4,4'-DDT	0.01
4,4'-DDE	0.05
4,4'-DDD	0.05
Dieldrin	0.01
Alpha-Endosulfan	0.02
Beta-Endosulfan	0.01
Endosulfan Sulfate	0.05
Endrin	0.01
Endrin Aldehyde	0.01
Heptachlor	0.01
Heptachlor Epoxide	0.01
PCB 1016	0.5
PCB 1221	0.5
PCB 1232	0.5
PCB 1242	0.5
PCB 1248	0.5
PCB 1254	0.5
PCB 1260	0.5
Toxaphene	0.5

Techniques:

GC - Gas Chromatography

GCMS - Gas Chromatography/Mass Spectrometry

HRGCMS - High Resolution Gas Chromatography/Mass Spectrometry (i.e., EPA 1613, 1624, or 1625)

LC - High Pressure Liquid Chromatography

FAA - Flame Atomic Absorption

GFAA - Graphite Furnace Atomic Absorption

HYDRIDE - Gaseous Hydride Atomic Absorption

CVAA - Cold Vapor Atomic Absorption

ICP - Inductively Coupled Plasma

ICPMS - Inductively Coupled Plasma/Mass Spectrometry

SPGFAA - Stabilized Platform Graphite Furnace Atomic Absorption (i.e., EPA 200.9)

DCP - Direct Current Plasma

COLOR - Colorimetric

⁵

The normal method-specific factor for these substances is 100, therefore, the lowest standard concentration in the calibration curve is equal to the above ML value for each substance multiplied by 100.

State of California
California Regional Water Quality Control Board
Santa Ana Region

Staff Report

June 19, 2015

ITEM: *9

SUBJECT: Issuance of Waste Discharge Requirements and Master Reclamation Permit for El Toro Water District, Water Recycling Plant. — Order No. R8-2015-0023, Orange County

DISCUSSION:

See the attached Fact Sheet.

RECOMMENDATIONS:

Adopt Order No. R8-2015-0023 as presented.

COMMENT SOLICITATION:

Comments were solicited from the discharger and the following agencies:

State Water Resources Control Board, Office of the Chief Counsel - David Rice
State Water Resources Control Board, DWQ – Annalisa Kihara
State Water Resources Control Board, DDDW, Santa Ana - Oliver Pacifico
California Regional Water Quality Control Board, San Diego Region, Brandi Outwin-Beals
South Orange County Wastewater Authority – Brennon Flahive
Irvine Ranch Water District – Lyndy Lewis
Orange County Water District – Jason Dadakis
Orange County Sanitation District – James D. Ruth
Orange County Health Care Agency – Larry Brennler
Orange County Public Works - Chris Crompton/Andy Ngo
El Toro Water District – Dennis Cafferty
Orange County Coastkeeper – Garry Brown